FROM SENSOR READINGS TO GAIT INSIGHTS

A practical project roadmap for analyzing real-world gait recordings collected with an ESP32 and GY-521 / MPU6050 accelerometer-gyroscope sensor.

Project aim

Turn raw movement-sensor readings into a reproducible data-science study: validate the data, visualize the signals, compare left and right legs within each participant, extract features, and explore whether simple statistical or machine-learning methods reveal useful patterns.

> **Recommended first milestone: build a trustworthy left-versus-right exploratory analysis and participant-level variability report before attempting predictive modelling.**

What this guide covers

- Exploratory data analysis and data-quality checks
- Left/right leg comparisons and paired statistics
- Signal processing, windowing, and feature extraction
- Clustering, anomaly detection, and carefully scoped ML experiments
- Visualizations, evaluation rules, limitations, and a practical work plan

Scope note: this is a project plan based on the described sensor setup and recordings. Confirm the actual number of participants, recording duration, sampling frequency, axis units, and file structure from your CSVs before reporting them as dataset facts.

1. Understand and validate the dataset

Start by confirming what each row represents and how recordings are organized. The MPU6050 typically provides three accelerometer axes and three gyroscope axes; exact units and scale depend on sensor configuration and the code used to read it.

- **Metadata:** participant ID, leg (left/right), trial or recording ID, timestamp, and any notes about the walking protocol.
- **Channels:** acceleration X/Y/Z and gyroscope X/Y/Z; document whether values are raw counts, g, m/s², degrees/s, or radians/s.
- **Quality checks:** missing values, duplicated timestamps, irregular sampling intervals, abrupt cut-offs, constant channels, and implausible spikes.
- **Sampling rate:** estimate it from timestamps rather than assuming the configured rate was achieved consistently.
- **Pairing:** verify that each left-leg recording is matched to the correct right-leg recording from the same participant and comparable protocol.

2. Exploratory data analysis (EDA)

Use EDA to learn what the sensor captured before selecting algorithms.

- Plot each of the six channels against time for representative recordings.
- Inspect the same channel across participants to see between-person variation.
- Check distributions, ranges, quantiles, and outliers.
- Compare signal magnitude and variability between recordings.
- Create a compact data-quality summary showing duration, sample count, missingness, and estimated sampling rate per file.

3. Left-leg versus right-leg analysis

Because both legs belong to the same participant, treat the comparison as paired. Avoid treating the left and right recordings as if they came from unrelated people.

- **Visual comparison:** overlay signals when timing and sensor placement make that comparison meaningful; otherwise use aligned small-multiple plots.
- **Time-domain features:** mean, standard deviation, variance, RMS, minimum, maximum, peak-to-peak range, and signal energy.
- **Frequency features:** dominant frequency or spectral energy, if recordings are long enough and sampling quality supports it.
- **Paired statistics:** calculate each participant's left-right difference, report its distribution and effect size, and use a paired test only when its assumptions are reasonably met.
- **Interpretation:** a difference may reflect placement, orientation, speed, or protocol variation—not necessarily biological asymmetry.

4. Signal processing and segmentation

Raw sensor readings may contain noise and orientation effects. Keep a copy of the raw data, then apply processing in a documented, reproducible pipeline.

- Check sensor orientation and whether gravity is present in the acceleration channels.
- Use a suitable filter only when justified by the signal and analysis goal; record the filter type and cut-off frequency.
- Normalize features only when appropriate, and fit any learned scaling using training data alone.
- Divide recordings into fixed-duration windows only after checking that the recordings contain enough samples and the windows suit the gait cycle.

5. Feature extraction

A feature table summarizes each window. A row might represent one participant, one leg, one recording, and one time window. Keep the identifiers so the source of every feature row remains traceable.

Feature family	Examples	Why it may help
Time domain	Mean, standard deviation, variance, min/max, peak-to-peak	Summarizes level and variation of movement signals.
Magnitude	RMS, vector magnitude, signal energy	Captures overall signal intensity; check units and gravity effects.
Frequency domain	Dominant frequency, spectral energy	Can describe repetitive patterns when sampling and duration are adequate.
Cross-channel	Correlation between axes; selected axis ratios	Explores relationships between sensor channels.

6. Statistical exploration and dimensionality reduction

- **Correlation matrix:** identify features that are strongly related.
- **PCA:** summarize correlated features and visualize broad patterns. Fit PCA on training data only if it is part of a predictive pipeline.
- **Participant-level summaries:** compare distributions across people while remembering that a small sample produces uncertain estimates.
- **Clustering:** try a method such as k-means or DBSCAN only after scaling suitable features and choosing a meaningful unit of analysis. Clusters require interpretation and validation; they are not automatically gait categories.

7. Machine-learning experiments: what is and is not supported

Idea	Possible approach	Important caveat
Unusual-window exploration	Isolation Forest or Local Outlier Factor	Flags statistical outliers, not medically abnormal gait.
Unsupervised grouping	Clustering extracted features	Small samples and correlated windows can produce unstable groups.
Left/right classification	SVM, KNN, Random Forest as a demonstration	May learn participant identity or recording setup rather than general leg differences.
Activity recognition	Classification after collecting labeled activities	Walking-only data cannot support reliable sitting/running/stairs classes.
TinyML	Compress/deploy a validated small model on ESP32	Do this only after defining a useful target and testing generalization.

8. Recommended visualizations

Visualization	What it shows	Priority
Six-channel time series	Raw acceleration and gyroscope signals against time.	First
Paired left/right plots	Within-person differences and recording consistency.	First
Feature box plots	Spread and potential outliers in features by leg and participant.	High
Correlation heatmap	Relationships among sensor axes and extracted features.	High
PCA scatter plot	A two-dimensional exploratory view of feature patterns.	Optional
Anomaly score plot	Windows flagged by an unsupervised method for follow-up inspection.	Optional

9. A practical work plan

Stage	Tasks	Deliverable
1. Validate	Inspect files, metadata, sampling, units, missingness, pairing.	Data-quality summary
2. Visualize	Plot channels and representative recordings; compare legs per participant.	EDA notebook and figures
3. Extract	Create window-level feature table with IDs preserved.	Feature CSV
4. Analyze	Summarize paired differences and participant variability.	Results table and plots
5. Explore	Optionally test clustering or anomaly detection and inspect flagged cases.	Exploratory findings
6. Communicate	Document methods, limitations, and reproducible code.	Report or dashboard

10. Evaluation and common pitfalls

- **Participant leakage:** never split windows randomly across train/test if windows from the same participant can appear on both sides. Keep all recordings and legs from a participant in one split.
- **Small sample size:** many windows from a few people do not equal many independent people. Report the participant count separately from the window count.
- **No ground-truth labels:** without reference labels, you cannot calculate meaningful supervised classification accuracy or prove clinical anomaly detection.
- **Sensor placement:** mounting position, orientation, walking speed, and protocol differences can dominate signal differences.
- **Clinical claims:** do not describe a flagged recording as pathological or diagnose a condition from this dataset.
- **Privacy and consent:** store participant IDs rather than names and ensure participants agreed to collection and use of their data.

11. Suggested project framing

Possible title: “Exploratory Computational Analysis of Wearable Inertial Sensor Data for Gait Pattern Characterization.”

Research question: “How consistently can simple time- and frequency-domain features summarize short gait recordings, and what within-participant differences are observed between left- and right-leg recordings?”

Expected contribution: a documented sensor-data pipeline, interpretable plots, a feature table, paired comparisons, and a transparent discussion of data limitations. This is a credible pilot analysis even if no predictive model is ultimately